

Rohithkumar Saravanan

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Professional Summary

AI, data and software graduate with an MSc in Artificial Intelligence and Machine Learning (University of Limerick, 2026, 2:1) and a B.Tech in Electronics and Computer Engineering (First Class Honours with Distinction). Builds practical solutions to real problems, from a roster-automation application for a private company to a full-stack AI-powered stock analysis platform with live data integration and 70+ automated tests. Brings a year of experience supporting databases and production environments for client accounts within a 24/7 support model, three Microsoft certifications in Fabric analytics, data engineering and AI-enabled SQL, and machine learning research recognised with a Springer publication, a Best Paper Award and a Best Project Award.

Education

Master of Science (MSc) in Artificial Intelligence and Machine Learning 2025 – 2026

University of Limerick – Second Class Honours, Grade 1 (2:1)

Bachelor of Technology (B.Tech) in Electronics and Computer Engineering 2021 – 2025

SRM Institute of Science and Technology, India – First Class Honours with Distinction

Technical Skills

- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, CNN/LSTM, Transfer Learning, Genetic Programming, TensorFlow, DEAP, Statistical Testing
- **Generative AI:** LLM Integration (Anthropic Claude API), RAG, Prompt Engineering
- **Data & Cloud:** Microsoft Fabric, Azure SQL, AWS, Data Pipelines (ETL), Power BI, pandas, Matplotlib
- **Databases:** SQL, PostgreSQL, SQLite, Redis, SQLAlchemy
- **Programming:** Python, TypeScript/JavaScript, SQL, MATLAB, C/C++, HTML/CSS
- **Web & APIs:** FastAPI, Next.js, React, Angular, Streamlit, REST APIs, JWT Authentication
- **Tools & Operations:** Git, GitHub, pytest, 24/7 Production Support, Environment Clones, Health Checks
- **Professional:** Problem-Solving, Communication, Teamwork, Leadership, Adaptability

Experience

IT Support Engineering Intern Aug 2024 – Aug 2025

Doyensys (India)

- **Globus Medical (client):** Provided full database support across production and non-production environments as part of a 24/7 support model.
- Carried out month-end activities, executed scripts on request, and handled user password resets for the Globus Medical account.
- **Petmate (client):** Performed environment clones and routine health checks, and maintained production, test and development environments with 24/7 support coverage.

Python Developer Intern Aug 2023 – Nov 2023

Shiash Info Solutions Private Limited (India)

- Developed Python-based models to estimate carbon emissions from industrial and household sources, applying data analysis to support environmental sustainability and industrial efficiency.

Cyber Security Intern (Industrial Training) Mar 2023 – Apr 2023

Industrial Training Program

- Completed industry-mentored training in threat analysis, system vulnerabilities and security best practices.

Projects

Work Scheduler – Roster Automation for Specialised Sterile Environments Ltd (SSE) Mar 2026

Private Company Project · Python, Streamlit, SQLAlchemy, SQLite, pandas, Matplotlib

- Built an end-to-end roster automation application for a cleaning contractor managing 30+ staff, about 50 recurring tasks and 4 cleanrooms across Morning, Evening and Night shifts, after observing supervisors lose hours daily to manual scheduling and absence cover.
- Implemented a greedy constraint solver that balances workload, respects per-machine certifications and training records, applies week-over-week rotation and surfaces coverage gaps, plus a same-day cover algorithm that reassigns an absent employee's tasks and auto-reverts the next day.

- Created a fairness-aware nightly cleanroom rotation using a 14-day assignment history, and shipped a multi-page Streamlit interface with PNG/PDF roster exports and an automated test suite.

Ledgerline – AI-Powered Stock Analysis Platform

Jun 2026

Personal Project · Python (FastAPI), TypeScript (Next.js/React), PostgreSQL, Redis, Anthropic Claude API

- Built a full-stack stock research platform with a FastAPI/PostgreSQL backend and Next.js/React frontend, integrating live market data from the Finnhub API behind a clean provider abstraction.
- Designed a transparent AI scoring model from 6 weighted technical and fundamental sub-scores, and a leak-free backtesting engine with a proper train/validation/test split.
- Integrated an LLM (Anthropic Claude) for natural-language score explanations; implemented JWT authentication, Redis caching and rate-limiting, and 70+ automated tests (pytest).

MSc Dissertation – Island-Model Evolution of Fuzzy Pattern Trees

2026

University of Limerick · Python, DEAP, Zuffy

- Investigated whether island-model genetic programming with migration improves the generalisation of interpretable Fuzzy Pattern Tree classifiers, running experiments on five benchmark datasets and evaluating results with Wilcoxon signed-rank tests across 240 parameter configurations.

Deep Learning Research – Medical Imaging and Biomedical Signals

2024 – 2025

SRM Institute of Science and Technology

- **Surface EMG Analysis for ALS (Best Project Award):** built Arduino-based signal-acquisition hardware and CNN, LSTM and CNN-LSTM classifiers, with a web interface for real-time prediction.
- **Pneumonia diagnosis from lung CT scans:** biorthogonal wavelet enhancement with a DenseNet classifier, achieving 99.20% accuracy (published in Springer); further models include a triplet-attention residual network (95.34% accuracy) and GoogLeNet transfer learning for lung cancer classification.

Certifications

- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600) – Sep 2026
- Microsoft Certified: Fabric Data Engineer Associate (DP-700) – Aug 2026
- Microsoft Certified: SQL AI Developer Associate (DP-800) – Jul 2026
- AWS Academy Graduate – AWS Academy Machine Learning Foundations
- Deloitte Data Analytics Job Simulation – Jul 2026
- **Online Courses:** Generative Deep Learning with TensorFlow, Advanced Computer Vision with TensorFlow, Supervised Machine Learning, Version Control with Git (Coursera); Advanced Diploma in Python Programming (CSC Institute)

Publications & Awards

- Biorthogonal Wavelet-Based Image Enhancement for Accurate Pneumonia Diagnosis from Lung CT Scans Using Deep Learning – Springer (AICTA 2024)
- Best Paper Award – All India Two Days Seminar, The Institution of Engineers (India), Jul 2024
- Best Project Award – Multidisciplinary Domain, SRM Institute of Science and Technology, Apr 2025
- 3rd Place – COMSPRO 2025 Project Contest, SRM Institute of Science and Technology, Apr 2025

Leadership

Committee Head – Directorate of Student Affairs (DSA)

Jun 2021 – May 2025

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